



humanovo

A search engine for the *unwritten* — turning the published biomedical literature into testable, mechanism-aware hypotheses, in minutes instead of weeks.

FOUNDER	STAGE	SECTOR	RAISING
Satvik Adyanthaya Founder	Pre-seed Live product	AI · Scientific tools B2B SaaS	\$1,000,000 Speedrun SAFE



THE PROBLEM

02 / 13 THE PROBLEM

The literature already contains the answers.
Nobody can read five million papers a year.

5M+

Academic & scientific papers published globally each year, 2026.

13K/day

Roughly 5,500–13,000 new papers published every single day.

~90%

Of clinical trials fail — most because the underlying hypothesis was wrong.
In many cases, contradicting evidence was already published before the trial began.

Sources: WordsRated / Scopus / Web of Science publication statistics 2025–2026; BIO / Biomedtracker on Phase I–III clinical-trial attrition; BMJ 2024 systematic review on preventable trial failures.

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THE PROBLEM



WHY NOW

03 / 13 WHY NOW

Three forces, in *sequence*, in the last eighteen months.

I. MODELS

Reason over full-text.

Frontier LLMs can finally hold an entire paper in context — methods, figures, supplementary. The unit of synthesis changed from abstracts to whole arguments.

II. ACCESS

Corpus is addressable.

Open-access mandates pushed millions of papers outside paywalls. For the first time, you can build a real product on top of the literature — with licensed full-text access on the roadmap.

III. METHOD

Adversarial grounding.

Multi-model adversarial pipelines catch the hallucinations that single-model RAG misses. Trustworthy generation finally became possible.

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WHY NOW



THE PRODUCT

04 / 13 THE PRODUCT

Plain English in. *Cited hypotheses out.*

A researcher types a hypothesis question in plain English. humanovo returns a ranked list of testable hypotheses — each grounded in cited papers, mechanism made explicit, every claim auditable back to source.

INPUTS

Natural-language questions across *treatment, prevention, biomarker, repurposing*, combination.

PIPELINE

A *twelve-stage adversarial* chain of frontier models, with dual-embedding grounding between every stage.

OUTPUTS

Ranked hypotheses with *full provenance* — sentence-level citations, mechanistic chain, translational roadmap.

EXAMPLE

“What mechanisms link GLP-1 agonists to neuroprotection in Parkinson’s?” → ranked hypotheses with cited papers and translational pathways, in minutes.

Live product. Demo available on request.

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THE PRODUCT



05 / 13 HOW IT WORKS

Twelve adversarial stages. Every claim *grounded twice*.

01-02 *Seed & Expand.* Claude Opus generates the initial hypothesis; Sonnet broadens scope.

03-05 *Evidence, Counter, Revise.* Cohere reviews the literature; Mistral attacks the hypothesis; o3-mini revises.

06-08 *Mechanism, Validate, Ground.* GPT-4.1 builds the mechanism; Sonnet cross-validates; Grok-4 grounds in three layers.

09-10 *Score & Refine.* Multi-dimensional confidence scoring, then fast refinement.

11-12 *Translate & Finalize.* Translational roadmap T0-T5; final synthesis with full citations.

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BETWEEN EVERY STAGE

Dual-embedding grounding — Cohere Embed v3 (1024d) and Azure text-embed-3-large (1536d) running in parallel.

Every claim is RAG-retrieved against *60+ data sources* — PubMed, ClinicalTrials.gov, openFDA, UniProt, Reactome, ChEMBL, AlphaFold — and ungrounded claims are gated out before the next stage runs.

A hypothesis that takes a postdoc *weeks* to assemble — sifting, reading, cross-referencing, arguing with their PI — is the unit of cost we are trying to compress.

I . Adversarial generation, not retrieval, is the unlock for <i>trustworthy</i> AI in science.	II . Mid-tier institutions are where bottlenecked human capital meets <i>willingness to pay</i> .	III . A platform for <i>hypotheses</i> compounds in a way a platform for papers cannot.
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VALIDATION

07 / 13 VALIDATION

In active use across *four UPMC departments*:
Radiation Oncology, Neuroimaging, Gastroenterology, Psychiatry.

“It does a *fantastic job* of summarizing potential hypotheses and proposing possible methods to solve it.”

— UPMC RESEARCHER
Unprompted, in-product feedback

ADVISORS

Roberto Colangelo
University of Rochester AIN

Dina Uzri
UR Ventures

Dr. Truptesh K
GI Director, UPMC

FOUNDER BACKGROUND

B.Tech Biotechnology · Research internship at Glenmark Pharmaceuticals — origin of the problem insight.

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VALIDATION

RECOGNITION

NYBPC Concept Award
Winner



08 / 13 PRICING

B2B from day one.
Free trial, then paid.

RESEARCHER

\$20 / mo

Individual scientists, postdocs, students

Limited monthly hypotheses*Caps designed for light use*

Core discovery modes & citation export

Top-ups available*Buy extra usage as needed*

TRIAL 3 hypotheses + 1 research-paper synthesis, free, before signup converts.

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PRICING

INSTITUTION

Custom

Universities, *pharma*, *research consortia*

Dedicated infrastructure & SLAs

Private corpus integration, SSO, audit logs

Solutions engineering & onboarding

PRICING

humanovo

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COMPETITION

Search. Summarize.

Adversarially generate.

CAPABILITY	PUBMED	CHATGPT	ELICIT	CO-SCIENTIST	HUMANOVO
Reasons over full-text papers	—	≈	✓	✓	✓
Adversarial multi-model pipeline	—	—	—	≈	✓
Dual-embedding claim grounding	—	—	—	—	✓
Generates novel hypotheses	—	≈	—	✓	✓
Auditable, sentence-level citations	✓	—	✓	≈	✓
Available to working PIs today	✓	✓	✓	—	✓

✓

meets

≈

partial

—

absent

Google AI Co-Scientist is closed beta — restricted to selected Google partner institutions.

humanovo is the only adversarial hypothesis engine available to working PIs today.



MARKET

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Where the *willingness to pay* actually lives.

BEACHHEAD

~12K

U.S. mid-tier academic medical centers & research universities — bottlenecked PIs, real budgets, no Big-Pharma AI internal stack.

ADJACENT

~6K

Biotech R&D teams & CROs running early-stage discovery — pricing aligned with seat-based research-tool spend.

HORIZON

Pharma

Top-20 pharma — discovery, repurposing, biomarker programs. Six-figure custom contracts; entered via institutional pull, not cold sales.

We are not pricing to compete with PubMed (free) or ChatGPT (consumer). We are pricing against a postdoc's time — and labs already pay \$60k+/yr for that.

Counts are public NIH / Carnegie / NSF institution rolls; see also AAMC member institutions and BIO biotech-employer reports. Pricing reflects published seat-based research-tool comps.

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MARKET



11 / 13 THE VISION
PHASE TWO — after the engine becomes infrastructure

A journal that *pays* the people it stands on.

PHASE ONE the hypothesis engine. PHASE TWO humanovo becomes a *repository*. Researchers and publishers register their work; when their hypotheses or findings are cited inside other humanovo-derived work, they earn royalties on the downstream value. Discovery, attribution, and compensation finally close the loop.

01 Publish Researchers and publishers register findings into the humanovo corpus with provenance.	→ 02 Discover The pipeline cites those findings when generating hypotheses for other researchers.	→ 03 Earn Original authors and publishers receive royalties on every downstream citation.	↩ 04 Reinvest Royalties fund the next study. Science compounds, instead of decays.
11 / 13	THE VISION		



THE ASK

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\$1M

Twelve months to land design partners, harden the pipeline, and prove pull — on a Speedrun standard SAFE.

USE OF FUNDS

~55%	Engineering	•
	Senior hires for the pipeline & grounding layer	
~25%	Compute & data	•
	Frontier-model inference, full-text licensing	
~15%	GTM	•
	Design-partner motion, conferences, content	
~5%	Reserve	•
	Legal, compliance, runway buffer	

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THE ASK



PURPOSE

IN CLOSING

*Every breakthrough treatment
starts with a hypothesis.
We are building the place where the **next**
ten thousand of them are born.*



humanovo.net

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THANK YOU